Dr. Wei Guo

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Academic Positions

Hong Kong University of Science and Technology — Sept. 2023 – Present
Postdoctoral Fellow. Hosts: Prof. Khaled B. Letaief and Prof. Jun Zhang — Hong Kong SAR, China

Education

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) — Jul. 2023
Ph.D., Computer and Information Engineering. Advisor: Prof. Chuan Huang — Shenzhen, China

University of Electronic Science and Technology of China (UESTC) — Jun. 2017
B.Eng., Communications Engineering — Chengdu, China

Academic Visits

Purple Mountain Laboratories — Sept. 2021 – Nov.2021
Visiting student. — Nanjing, China

The Future Network of Intelligence Institute (FNii), CUHK-Shenzhen — Jun. 2019 – Dec.2019
Visiting student. Advisors: Prof. Chuan Huang and Prof. Shuguang Cui — Shenzhen, China

Shenzhen Reserach Institute of Big Data (SRIBD) — Feb. 2017 – Jun. 2017
Visiting student. Advisor: Prof. Shuguang Cui — Shenzhen, China

Current Research Interests

Wireless Communications, Generative AI, Integrated Communications and AI

Publications (✉ denotes correspondence)

Journals:

- Y. Wang, **W. Guo**✉, C. Huang✉, G. Chen, Y. Zhang, and S. Cui, "Point Cloud Environment-Based Channel Knowledge Map Construction," *IEEE Trans. Cogn. Commun. Netw.*, vol. 12, pp. 9263-9278, 2026.
- J. Tong, **W. Guo**, J. Shao, Q. Wu, Z. Li, Z. Lin, and J. Zhang, "WirelessAgent: Large Language Model Agents for Intelligent Wireless Networks," *China Commun.*, vol. 23, no. 3, pp. 265-285, Mar. 2026.
- T. Liao, **W. Guo**✉, H. He, S. Song, J. Zhang, and K. B. Letaief, "Joint Beamforming and Antenna Position Optimization for Fluid Antenna Assisted MU-MIMO Networks," *IEEE J. Sel. Areas Commun.* (JSAC), vol. 44, pp. 1209-1226, 2026.
- J. Shao, J. Tong, Q. Wu, **W. Guo**, Z. Li, Z. Lin, and J. Zhang, "WirelessLLM: Empowering Large Language Models Towards Wireless Intelligence," in *J. Commun. Inf. Netw.* (JCIN), vol. 9, no. 2, pp. 99-112, Jun. 2024.
- **W. Guo**, C. Huang, X. Qin, L. Yang, and W. Zhang, "Dynamic Clustering and Power Control for Two-Tier Wireless Federated Learning," in *IEEE Trans. Wireless Commun.* (TWC), vol. 23, no. 2, pp. 1356-1371, Feb. 2024.

- **W. Guo**, R. Li, C. Huang, X. Qin, K. Shen, and W. Zhang, "Joint Device Selection and Power Control for Wireless Federated Learning," in *IEEE J. Sel. Areas Commun.* (JSAC), vol. 40, no. 8, pp. 2395-2410, Aug. 2022.
- **W. Guo**, H. Zhang, and C. Huang, "Energy Efficiency of Two-Way Communications under Various Duplex Modes," in *IEEE Internet Things J.* (IoT-J), vol. 8, no. 3, pp. 1921-1933, Feb. 2021.

Magazines:

- B. Huang, Y. Wang, **W. Guo**✉, Z. Guo, D. Wu, R. Li, D. Liu, W. Lan, C. Huang✉, and S. Cui, "Building Low-Altitude Communication Networks: A Digital Twin-Based Optimization Framework," *IEEE Wireless Commun. Mag.*, early access, 2026.
- Y. Wang, C. Huang✉, S. Zhang, G. Chen, **W. Guo**✉, S. Lan, L. Xu, X. Cheng, X. Tang, and S. Cui, "Towards Precise Channel Knowledge Map: Exploiting Environmental Information from 2D Visuals to 3D Point Clouds," *IEEE Commun. Mag.*, early access, 2026.

Conference Proceedings:

- Y. Bian, **W. Guo**, Z. Wang, S. Song, J. Zhang, and K. B. Letaief, "GSBF: Gaussian Splatting for Environment-Aware Beamforming," in *Proc. IEEE Glob. Commun. Conf. (Globecom)*, Dec. 2026.
- S. Xie, **W. Guo**, S. Song, J. Zhang, Y.-J. Zhang, and K. B. Letaief, "Hybrid Precision Task-Oriented Communication for OFDM-Based Edge Inference Systems," in *Proc. IEEE Mediterranean Conf. Commun. Netw. (MeditCom)*, Jul. 2026.
- T. Liao, **W. Guo**, J. Qian, S. Song, J. Zhang, and K. B. Letaief, "Rethinking Mutual Coupling in Movable Antenna MIMO Systems," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2026.
- R. Li, Z. Xu, **W. Guo**, and Y.-J. Zhang, "Divergence Meets Dispersion: Efficient Wideband Beam Training for XL-MIMO," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2026. **Best Paper Award**
- Y. Bian, **W. Guo**, J. Yang, S. Song, J. Zhang, S. Jin, and K. B. Letaief, "Multi-modal Data Driven Virtual Base Station Construction for Massive MIMO Beam Alignment," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2026.
- E. Tang, **W. Guo**, H. He, S. Song, J. Zhang, and K. B. Letaief, "Accurate and Fast Channel Estimation for Fluid Antenna Systems with Diffusion Models," in *Proc. IEEE Glob. Commun. Conf. (Globecom)*, pp. 2066-2071, Dec. 2025.
- T. Liao, **W. Guo**, H. He, S. Song, J. Zhang, and K. B. Letaief, "Fluid Antenna-Assisted MU-MIMO Systems with Decentralized Baseband Processing," in *Proc. IEEE Glob. Commun. Conf. (Globecom)*, pp. 2372-2377, Dec. 2025.
- **W. Guo**, M. He, C. Huang, H. He, S. Song, J. Zhang, and K. B. Letaief, "Model-driven Deep Learning for Distributed Detection in WSNs with Binary Quantization," in *Proc. IEEE Mediterranean Conf. Commun. Netw. (MeditCom)*, Jul. 2024.
- **W. Guo**, R. Li, C. Huang, X. Qin, K. Shen, and W. Zhang, "Optimized Device Selection and Power Control for Wireless Federated Learning," in *Proc. IEEE Glob. Commun. Conf. (Globecom)*, Dec. 2022.
- **W. Guo**, C. Huang, X. Qin, L. Yang, and W. Zhang, "Dynamic Clustering and Power Control for Two-Tier Over-the-Air Federated Learning," in *Proc. IEEE/CIC Int. Conf. Commun. China (ICCC)*, Aug. 2022.

- **W. Guo** and C. Huang, "Energy Efficiency of Full-Duplex Two-Way Channels," in *Proc. Glob. Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2019.

Patents:

- CN patent: "一种面向无线联邦学习的用户协同传输方法", 黄川, 崔曙光, **郭玮**. 专利申请号: 202211032064.0, 2022.

Preprints:

- **W. Guo**, Y. Zhang, X. Ge, and J. Zhang, "Equal Path Cost, Unequal Output Effects: Understanding Perturbation Propagation in Diffusion Models," arXiv: 2610.11380, 2026.
- Y. Bian, K. Zhang, **W. Guo**, Z. Wang, S. Song, J. Zhang, and K. B. Letaief, "AIMS: An Agentic AI Framework for Sim-to-Real Multi-Modal ISAC," arXiv: 2609.39964, 2026.
- S. Xie, **W. Guo**✉, S. Song, J. Zhang, Y.-J. Zhang, K. B. Letaief, "Task-Oriented Communication with Hybrid-Precision Models", arXiv: 2607.16766, 2026, submitted to *IEEE Trans. Mobile Comput.*
- Y. Bian, **W. Guo**, J. Yang, S. Song, J. Zhang, S. Jin, and K. B. Letaief, "Multi-Modal Environment-Aware Beam Management for Massive MIMO: A Geometry-Driven Virtual Base Station Framework," arXiv: 2606.26567, 2026, currently under **major revision**.
- E. Tang, **W. Guo**✉, H. He, S. Song, J. Zhang, K. B. Letaief, "A Unified Two-Stage Generative Diffusion Framework for Channel Estimation and Port Selection in Multiuser MIMO-FAS", arXiv: 2605.29679, 2026, submitted to *IEEE Trans. Wireless Commun.*, currently under **major revision**.
- T. Liao, **W. Guo**✉, J. Qian, Z. Wang, S. Song, J. Zhang, and K. B. Letaief, "Rethinking Mutual Coupling in Movable Antenna MIMO Systems: Modeling and Optimization," arXiv: 2604.26282, 2026, submitted to *IEEE Trans. Wireless Commun.*
- J. Tong, Z. Li, F. Liu, **W. Guo**, and J. Zhang, "WirelessAgent++: Automated Agentic Workflow Design and Benchmarking for Wireless Networks," arXiv: 2603.00501, 2026, submitted to *IEEE Trans. Wireless Commun.*, currently under **major revision**.

Awards

IEEE WCNC Best Paper Award	2026
Exemplary Reviewer of IEEE Wireless Communications Letters	2020

Teachings

Teaching assistant at CUHK-Shenzhen:

EIE4006 Performances Evaluation of Communication Networks	Sum' 2020-21
CIE6126 Performances Analysis of Computer and Communication Networks	Sum' 2020-21
EIE3001 Signals and Systems	Spr' 2021

Academic Services

Conference Reviewers:

- IEEE Global Communications Conference (Globecom): 2019-2025
- IEEE/CIC International Conference on Communications in China (ICCC): 2019-2025
- IEEE Wireless Communications and Networking Conference (WCNC), 2024.
- IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2024.

Journal Reviewers:

- IEEE Transactions on Wireless Communications
- IEEE Transactions on Communications
- IEEE Journal on Selected Areas in Information Theory
- IEEE Wireless Communications Letters
- IEEE Systems Journal
- IEEE Transactions on Machine Learning in Communications and Networking
- IEEE Internet of Things Magazine
- China Communications
- Journal of Communications and Information Networks
- ...

Presentations

2024 IEEE MeditCom, Marid, Spain, Oral presentation
2022 IEEE Globecom, Rio de Janeiro, Brazil, virtual presentation
2022 IEEE/CIC ICCC, Foshan, China, Oral presentation
2019 IEEE GlobalSIP, Ottawa, Canada, Oral presentation